

NPTEL Course on
"Computer System Optimization for ML"

Lecture #37

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Agenda

- Checkpointing (contd.)
- Graph Neural Network

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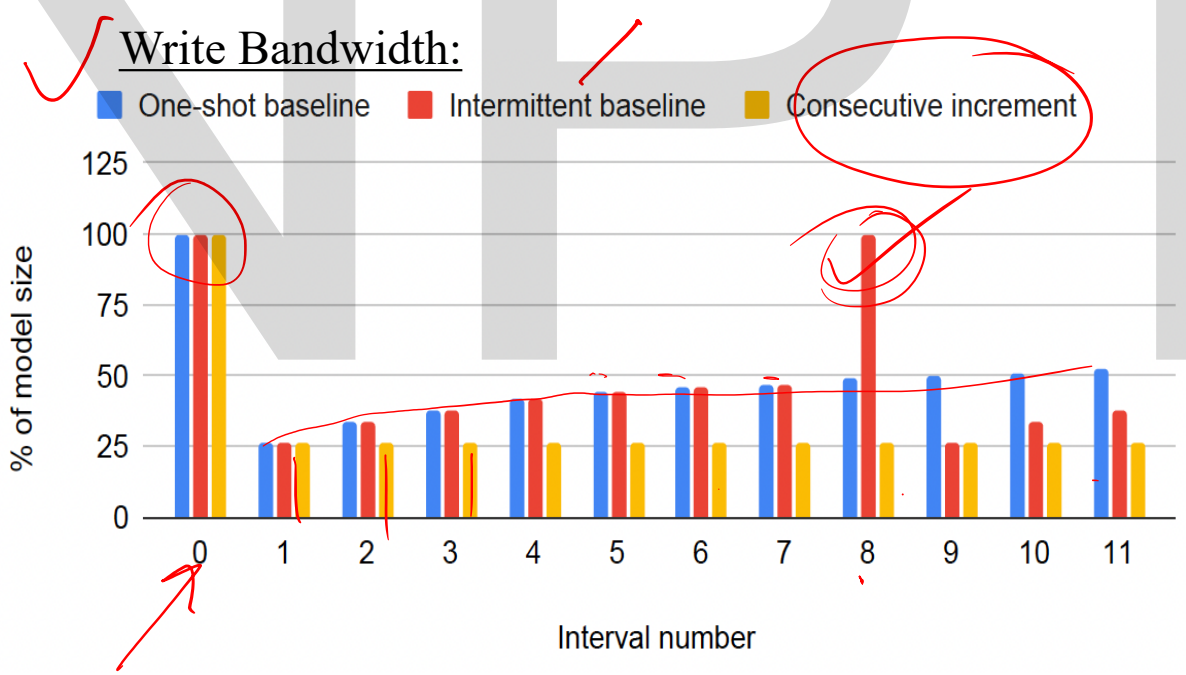
Approaches for Differential Checkpointing

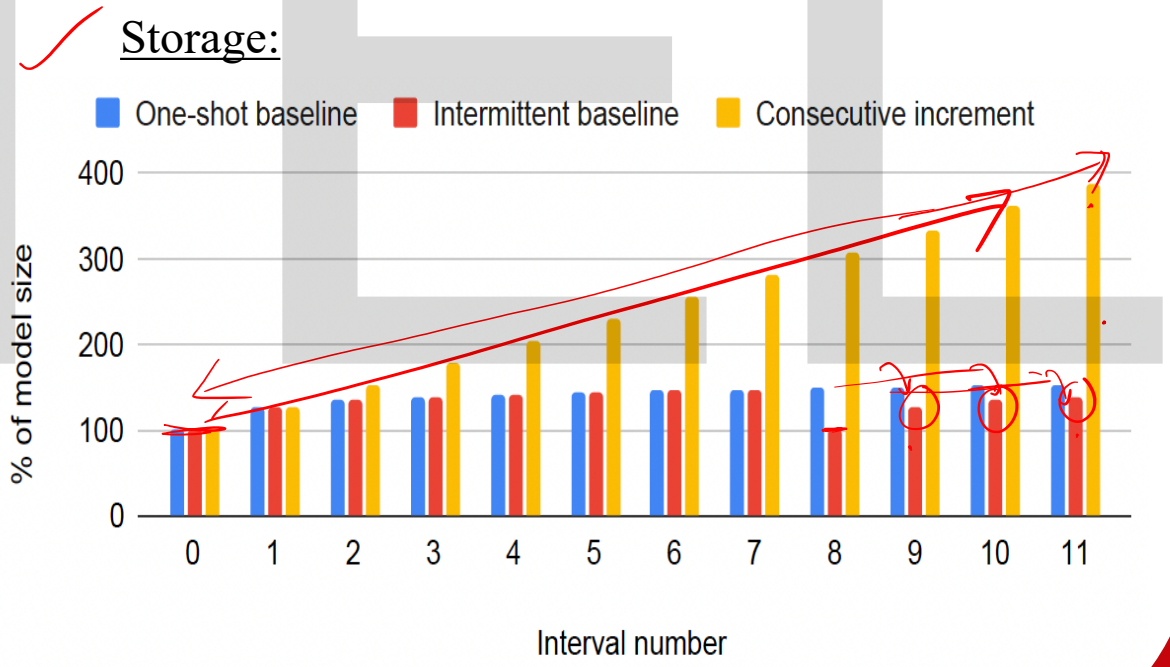
- One-Shot Differential Checkpoint
- Consecutive Incremental Checkpoint
  - Store changes since last checkpoint
- **Intermittent Differential Checkpoint**
  - **Take a full snapshot of model at regular interval so that subsequent differentials are small enough**
  - **At the end of each checkpoint interval, it estimates the expected cumulative size of future checkpoints if another differential checkpoint is taken, compared with the total expected size if a full checkpoint is taken**

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Approaches for Differential Checkpointing

- One-Shot Differential Checkpoint
- Consecutive Incremental Checkpoint
- Intermittent Differential Checkpoint



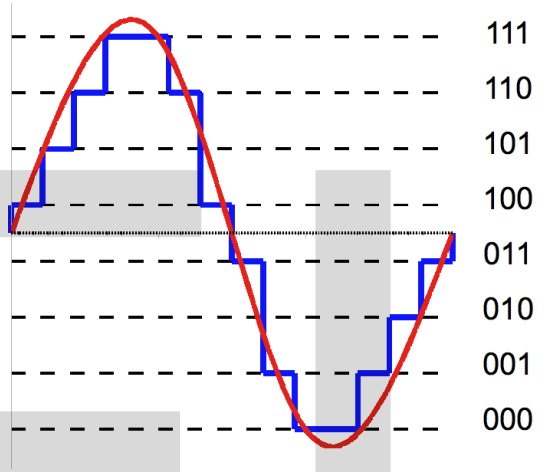


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Checkpoint Quantization

- **Compress checkpoint without degrading training accuracy**
- **Approaches**

…	..	…	…	…
0.21	-0.31	0.03	0.01	-0.05
…	..	…	…	…
…	..	…	…	…



Uniform:



(-0.31) () (-0.05, 0.03, 0.01) (0.21)

Non-Uniform:



(-0.31) (-0.05) (0.03, 0.01) (0.21)

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Many More Ideas

- Logging based checkpointing
- Parallel recovery for multi-GPU training
- Using PM for checkpointing

Read :

Swift : Expedited Failure Recovery for Large-scale DNN Training [Arxiv Feb'23]

GPM: Leveraging Persistent Memory from a GPU [ASPLOS '22]

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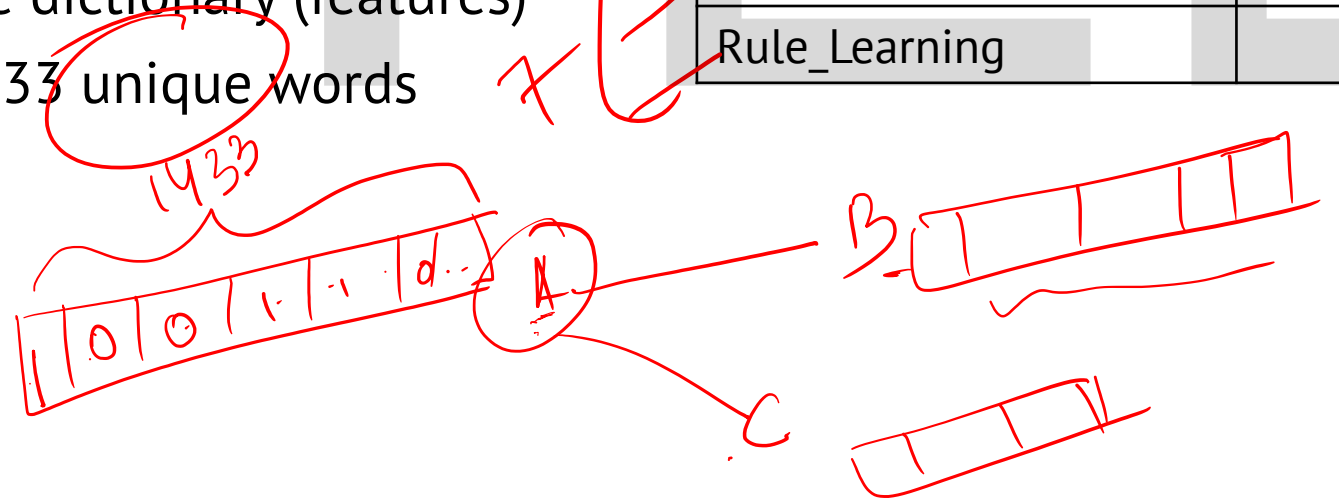
- Checkpointing (contd.)
- **Graph Neural Network**

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Graph Neural Network

- **Several estimation task can be represented as graph**
- **E.g. Cora dataset**
  - 2708 scientific publications
  - Classified into one of seven classes
  - The network consists of 5429 links (citations)
  - Each publication is represented by a 0/1-valued word vector indicating absence/presence of the corresponding word from the dictionary (features)
  - The dictionary consists of 1433 unique words
- **Task: to predict the subject**

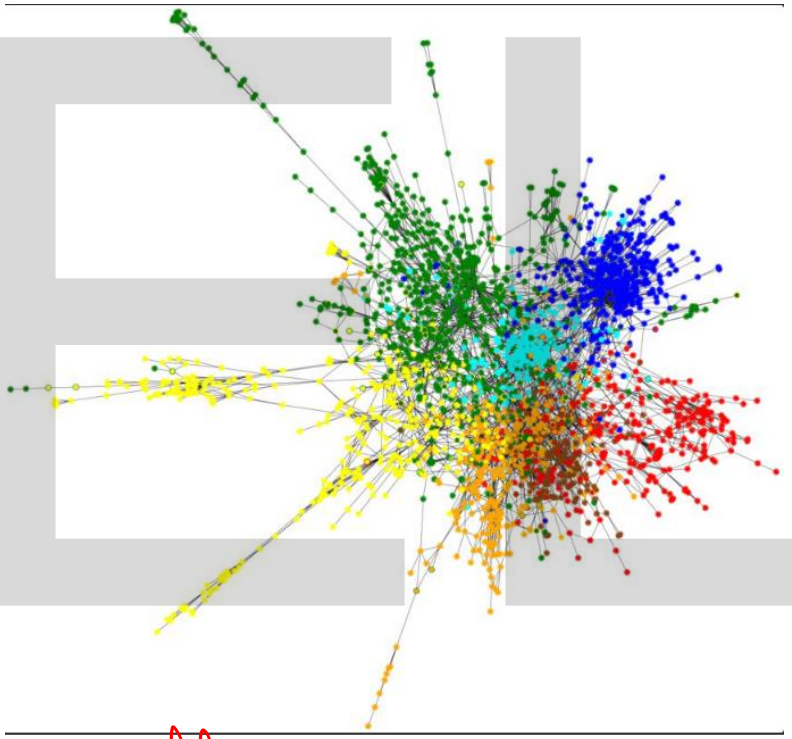
subject	# publications
Neural_Networks	818
Probabilistic_Methods	426
Genetic_Algorithms	418
Theory	351
Case_Based	298
Reinforcement_Learning	217
Rule_Learning	180



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Representation of Cora Dataset

- 2708 scientific publications
- Classified into one of seven classes
  - Each colour denotes a class
- The network consists of 5429 links (citations)
- Each publication is represented by a 0/1-valued word vector indicating absence/presence of the corresponding word from the dictionary (features)
- The dictionary consists of 1433 unique words



7 diff. colors

Another Example: REDDIT-Binary

- A graph corresponding to online discussion on Reddit
- Nodes represent users
- If there is an edge between them if at least one of them respond to the other's comment
- Classification
  - Question/Answer-based community
  - Discussion-based community
- Statistics
  - Nodes: 232, 965
  - Edges: 114, 615, 892
  - Node feature size: 602

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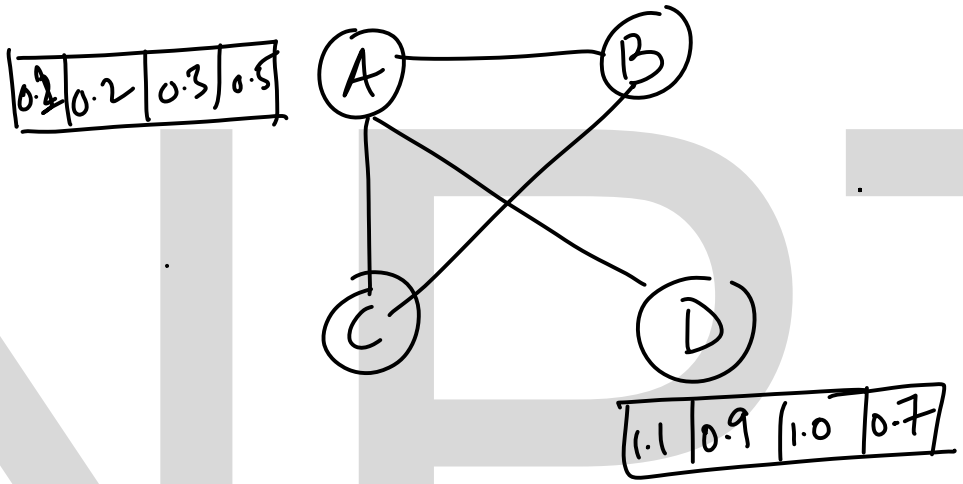
GNN Training and Inference

- **Chalk and Talk**

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Inference, Training.

MLP



$G(V, E)$

Adjacency Matrix

	A	B	C	D
A	0	1	1	1
B	1	0	1	0
C	1	1	0	0
D	1	0	0	0

GNN inference

1. Aggregation
2. Combination

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Iteration-1

1. Aggregation — ⓓ collecting all features from the neighbor node

ⓓ will get features of ⓐ

ⓓ	1.1	0.9	1.0	0.7
ⓐ	0.1	0.2	0.3	0.5

f_1^D

f_1^A

Compute a new feature set

$\bar{f}_D = g(f_1^D, f_1^A)$

$g = '+'$

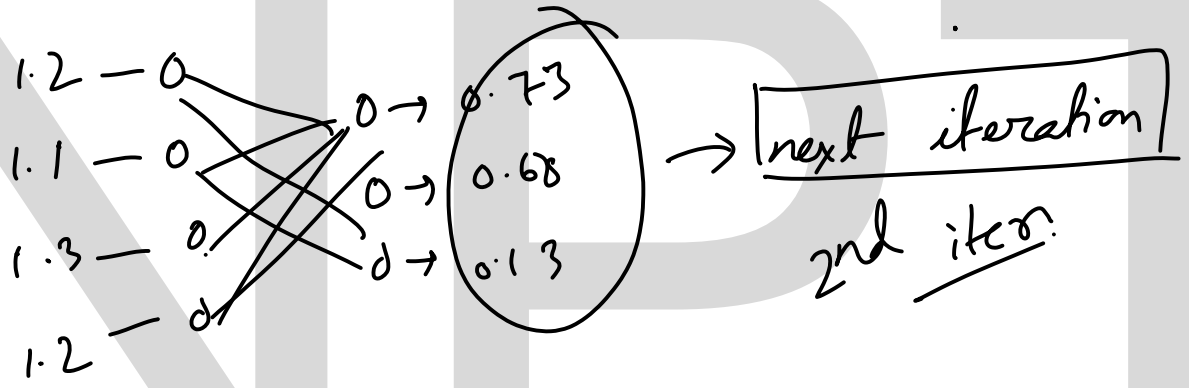
$\bar{f}_D$ =

1.2	1.1	1.3	1.2

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② Combination

Each node of GNN consists of a shallow MLP.



Two imp info

1. The MLP structure as well as the abs val. of the weights are same for all nodes.
2. For each iteration, there is a separate MLP.

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)